

S5.4 Transformations of the Sinusoidal Functions

October 25, 2023 8:44 PM



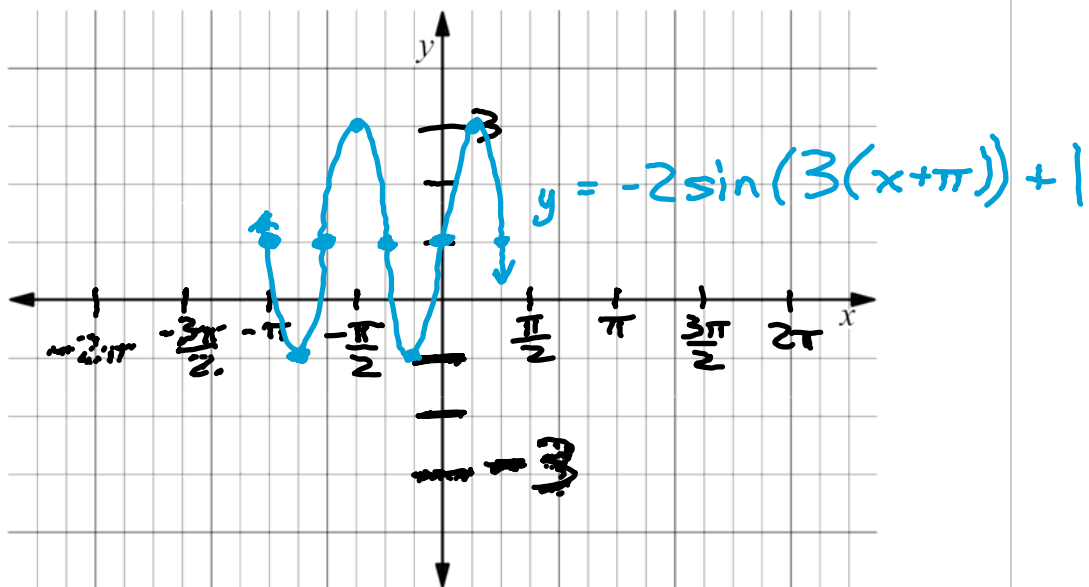
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(5.4) Transformations of the Sinusoidal Functions

We are going to apply similar transformations to trigonometric functions that we did to other functions only that we are using Radian measure.

Graph $y = -2 \sin(3(x + \pi)) + 1$ for two cycles.

$$\begin{aligned} (x, y) &\rightarrow \left(\frac{x}{3} - \pi, -2y + 1 \right) \\ (0, 0) &\rightarrow (-\pi, 1) \\ \left(\frac{\pi}{2}, 1 \right) &\rightarrow \left(-\frac{3\pi}{6}, -1 \right) \\ (\pi, 0) &\rightarrow \left(-\frac{2\pi}{3}, 1 \right) \\ \left(\frac{3\pi}{2}, -1 \right) &\rightarrow \left(-\frac{\pi}{2}, 3 \right) \\ (2\pi, 0) &\rightarrow \left(-\frac{\pi}{3}, 1 \right) \end{aligned}$$



Ex. 1 Determine the amplitude, period, phase shift and axis of curve for:

$$y = \underline{3} \cos \left(\underline{2\theta} + \frac{\pi}{3} \right) \underline{-5} \rightarrow y = 3 \cos \left(2 \left(\theta + \frac{\pi}{6} \right) \right) - 5$$

$$\text{Amplitude} = 3$$

$$\text{Period} = \frac{2\pi}{2} \\ = \pi$$

$$\text{Phase Shift} \\ = \frac{\pi}{6} \text{ (left)}$$

$$\text{Axis of Curve} \\ y = -5$$